

2. Use an abstract factory pattern to design a set of skin libraries. The skin library has Spring and Summer styles. Spring style uses green buttons, text boxes, and combo boxes, and Summer style uses blue components. Write a SkinFactory interface to create the above two styles of skins. Write a Test class to use these two skins. When each component is created, it needs to display the creation information, such as "GreenBorderTextBox created!".

BlueButtons

```
package experiment5_2;

class BlueButtons implements Buttons{

    @Override

    public void produceButtons() {

        System.out.println("Blue Button created!");

    }

}
```

BlueComboBoxes

```
package experiment5_2;

public class BlueComboBoxes implements ComboBoxes{

    @Override

    public void produceComboBoxes() {

        System.out.println("Blue Combo Boxes created!");

    }

}
```

BlueTextBoxes

```
package experiment5_2;

public class BlueTextBoxes implements TextBoxes{

    @Override

    public void produceTextBoxes() {

        System.out.println("Blue Text Boxes created!");

    }

}
```

Buttons

```
package experiment5_2;

interface Buttons {

    void produceButtons();

}
```

ComboBoxes

```
package experiment5_2;

interface ComboBoxes {

    void produceComboBoxes();

}
```

GreenButtons

```
package experiment5_2;

public class GreenButtons implements Buttons{

    @Override

    public void produceButtons() {

        System.out.println("Green Button created!");

    }

}
```

GreenComboBoxes

```
package experiment5_2;

public class GreenComboBoxes implements ComboBoxes{

    @Override

    public void produceComboBoxes() {

        System.out.println("Green Combo Boxes created!");

    }

}
```

GreenTextBoxes

```
package experiment5_2;

public class GreenTextBoxes implements TextBoxes{

    @Override
```

```

    public void produceTextBoxes() {
        System.out.println("Green Text Boxes created!");
    }
}

```

SkinFactory

```

package experiment5_2;

interface SkinFactory {
    Buttons createButtons();
    ComboBoxes createComboBoxes();
    TextBoxes createTextBoxes();
}

```

SpringSkinFactory

```

package experiment5_2;

public class SpringSkinFactory implements SkinFactory{
    @Override
    public Buttons createButtons() {
        return new GreenButtons();
    }

    @Override

```

```
public ComboBoxes createComboBoxes() {  
    return new GreenComboBoxes();  
}  
  
@Override  
public TextBoxes createTextBoxes() {  
    return new GreenTextBoxes();  
}  
}
```

SummerSkinFactory

```
package experiment5_2;  
  
public class SummerSkinFactory implements SkinFactory{  
    @Override  
    public Buttons createButtons() {  
        return new BlueButtons();  
    }  
  
    @Override  
    public ComboBoxes createComboBoxes() {  
        return new BlueComboBoxes();  
    }  
}
```

```
@Override  
  
public TextBoxes createTextBoxes() {  
    return new BlueTextBoxes();  
}  
}
```

Test

```
package experiment5_2;  
  
public class Test {  
    public static void main(String[] args) {  
        Buttons buttons1, buttons2;  
        ComboBoxes comboBoxes1, comboBoxes2;  
        TextBoxes textBoxes1, textBoxes2;  
        SkinFactory skinFactory1, skinFactory2;  
  
        System.out.println("Spring Style");  
        skinFactory1 = new SpringSkinFactory();  
        buttons1 = skinFactory1.createButtons();  
        buttons1.produceButtons();  
        comboBoxes1 = skinFactory1.createComboBoxes();  
        comboBoxes1.produceComboBoxes();  
    }  
}
```

```

        textBoxes1 = skinFactory1.createTextBoxes();
        textBoxes1.produceTextBoxes();

        System.out.println();

        System.out.println("Summer Style");
        skinFactory2 = new SummerSkinFactory();
        buttons2 = skinFactory2.createButtons();
        buttons2.produceButtons();
        comboBoxes2 = skinFactory2.createComboBoxes();
        comboBoxes2.produceComboBoxes();
        textBoxes2 = skinFactory2.createTextBoxes();
        textBoxes2.produceTextBoxes();

    }
}

```

TextBoxes

```

package experiment5_2;

interface TextBoxes {
    void produceTextBoxes();
}

```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ
Spring Style
Green Button created!
Green Combo Boxes created!
Green Text Boxes created!

Summer Style
Blue Button created!
Blue Combo Boxes created!
Blue Text Boxes created!

Process finished with exit code 0
```